



Hospital Patient Management System

Project Overview

হাসপাতাল রোগী ব্যবস্থাপনা সিস্টেম

Field	Details
Project Name	Hospital Patient Management System (Hospital PMS)
Type	Full-Stack Web Application
Country	Bangladesh BD
Tech Stack	Django 6.0.5 + React 19 + PostgreSQL 15
Document Type	Project Overview
Version	1.0 — May 2026
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Developed for Bangladesh Healthcare

1. What is Hospital PMS?

Hospital PMS (Hospital Patient Management System) is a complete, full-stack web-based application designed to digitize and manage the day-to-day operations of a multi-department hospital in Bangladesh.

The system eliminates paper-based hospital processes by providing a modern, browser-based interface that hospital staff can access from any computer on the hospital network. It covers the complete patient journey — from initial registration at the front desk, through doctor consultations and lab tests, to billing and payment collection.

Built using industry-standard technologies — Django REST Framework on the backend and React on the frontend — the system is secure, fast, and designed specifically for the Bangladesh healthcare environment, supporting local payment methods like bKash, Nagad, and Rocket.

2. Key Highlights

Highlight	Detail
 Secure Authentication	JWT-based login with 5 user roles and automatic token expiry
 Patient Auto-ID	Patients receive unique IDs like P-2026-0001 automatically
 Smart Scheduling	Book and manage appointments across multiple departments
 Bangladesh Payments	Accept bKash, Nagad, Rocket, Cash, and Card payments
 Digital Prescriptions	Doctors write digital prescriptions with medicine details
 Lab Workflow	Full sample-to-result tracking for lab tests
 Print Invoices	Generate professional PDF invoices with one click
 Live Analytics	Real-time reports on revenue, patients, and departments
BD Bengali Language Support	Interface supports both Bengali and English text
 PostgreSQL Database	Reliable relational database with 15+ tables

3. System Modules

The Hospital PMS consists of 7 fully functional modules:

Module 1 — Authentication

Every user must log in with a username and password. The system uses JWT (JSON Web Token) technology — a secure industry standard — to verify identity. Once logged in, users see only the features appropriate for their role.

Role	What They Can Do
Admin	Full access to all modules, manage users and settings
Doctor	Write prescriptions, order lab tests, view patient history
Receptionist	Register patients, book appointments, manage billing
Nurse	View patients and appointments
Lab Technician	Process lab orders and record test results

Module 2 — Patient Management

The patient module is the heart of the system. Every person who visits the hospital is registered here. The system automatically generates a unique Patient ID in the format P-YYYY-NNNN (for example, P-2026-0001) which is used throughout the system.

- Register new patients with full personal details
- Auto-generate unique Patient IDs
- Record blood group, NID number, allergies, and chronic conditions
- Search patients instantly by name, phone, patient ID, or NID
- View and update patient records
- Soft-delete (deactivate) patients without losing data

Module 3 — Appointments

Hospital staff can book and manage patient appointments across all departments. Each appointment is linked to a specific doctor and department, and its status is tracked from scheduling to completion.

- Book appointments for any registered patient
- Select department (Medicine, Cardiology, Gynecology, Surgery, Pediatrics, etc.)
- Select available doctor for that department
- Set appointment date, time, and type (OPD / Follow-up / Emergency)
- Track appointment status: Scheduled → Confirmed → In Progress → Completed
- View today's appointments with a dedicated filter
- Filter appointments by date and status

Module 4 — Billing & Payments

The billing module handles all financial transactions in the hospital. It generates professional invoices and supports multiple payment methods — including Bangladesh's most popular mobile banking services.

- Create invoices with multiple service line items
- Auto-calculate subtotal, discount, tax, and total amount

- Accept payments via Cash, bKash, Nagad, Rocket, and Credit/Debit Card
- Record bKash/Nagad transaction IDs for audit purposes
- Track invoice status: Pending, Paid, Partial, Cancelled
- View revenue summary: total collected and pending amounts
- Print professional PDF invoices with one click

Module 5 — Prescriptions

Doctors can write fully digital prescriptions for their patients. Each prescription records the diagnosis, medical advice, prescribed medicines with dosage instructions, and a follow-up date.

- Write prescriptions linked to patient and doctor
- Record diagnosis and medical advice
- Add multiple medicines with: name, dosage, frequency, duration, and instructions
- Frequency options: Once daily, Twice daily, Three times daily, When needed, etc.
- Set follow-up appointment date
- View full prescription details in a clean modal popup

Module 6 — Lab & Diagnostics

The lab module manages the complete lifecycle of diagnostic test orders — from the doctor ordering the tests, through sample collection and processing, to final result delivery.

- Maintain a catalogue of available lab tests with prices (CBC, ECG, X-Ray, etc.)
- Doctors order one or more tests for patients
- Lab order numbers auto-generated (LAB-2026-0001 format)
- Track order through stages: Ordered → Sample Collected → In Progress → Completed
- Record test results and flag abnormal values
- Filter orders by status

Module 7 — Reports & Analytics

The reports module gives hospital management a real-time overview of all hospital activity. All data is fetched live from the database whenever the page is opened or refreshed.

- Total patients registered
- Total appointments booked
- Total revenue collected (from paid invoices)
- Pending payment amounts
- Total invoices, prescriptions, and lab orders
- Recent patient list with district and blood group
- Recent invoice list with status
- Department overview cards (Medicine, Cardiology, Surgery, etc.)

4. Technology Used

Layer	Technology	Purpose
Backend API	Django 6.0.5 + Django REST Framework	Handles all data and business logic
Database	PostgreSQL 15	Stores all hospital data securely
Authentication	JWT (JSON Web Tokens)	Secure login without sessions
Frontend UI	React 19 + Vite	Fast, interactive user interface
Styling	Tailwind CSS v4	Clean, professional appearance
State Management	Redux Toolkit	Manages login state across pages
HTTP Requests	Axios	Connects frontend to backend API
Payments	SSLCommerz	bKash, Nagad, Rocket integration
SMS	SSL Wireless Bangladesh	Appointment reminder messages
Dev Environment	PyCharm + pgAdmin 4	Development and DB management

5. Project Statistics

Metric	Value
Total Modules Completed	7
Django Backend Apps	7 (authentication, patients, appointments, billing, prescriptions, lab, + config)
Database Tables	15+
REST API Endpoints	20+
React Frontend Pages	9 (Login, Dashboard, Patients, Appointments, Billing, Prescriptions, Lab, Reports)
User Roles	5 (Admin, Doctor, Receptionist, Nurse, Lab Technician)
Payment Methods Supported	5 (Cash, bKash, Nagad, Rocket, Card)
Language Support	English + Bengali (বাংলা)
Timezone	Asia/Dhaka (UTC+6)
Auto-ID Fields	3 (Patient ID, Invoice Number, Lab Order Number)

6. How to Access the System

Action	Command / URL
Start backend (Terminal 1)	cd server → venv\Scripts\activate → python manage.py runserver
Start frontend (Terminal 2)	cd client → npm run dev
Open in browser	http://localhost:5173
Admin panel	http://127.0.0.1:8000/admin
Login (username)	admin
Login (password)	1234 (change in production)

7. Summary

The Hospital Patient Management System is a fully functional, production-ready web application that covers the core operational needs of a modern hospital in Bangladesh. Starting from zero, the entire system was built step by step — including the database design, REST API development, user interface, payment integration, and reporting.

The system demonstrates a complete full-stack development workflow using industry-standard tools and frameworks. It is a strong foundation that can be extended in the future with additional features such as mobile application support, insurance claim processing, ward/bed management, or integration with medical devices.